

Benjamin Stahl

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EDUCATION

Lehigh University, Bethlehem, PA — *B.S. Computer Science & Business, Minor in Data Science*

GPA: 3.52 | Expected Dec 2026 | Study Abroad: Barcelona, Spain (Summer 2025)

Relevant Coursework: Computer Architecture, Systems Software, Operating Systems, Machine Learning, Data Structures & Algorithms

UVM Training: Sunburst Design SystemVerilog UVM Verification & UVM RAL (Cliff Cummings), September 2026

TECHNICAL SKILLS

- **Verification:** UVM (sequences, TLM, scoreboards, factory, RAL), constrained-random, functional coverage (covergroups, crosses, URG), SVA, DPI (C++↔SV), cocotb
- **EDA Tools:** VCS, Verdi, Verilator, Coverity, Slang, LSF
- **Languages:** SystemVerilog, C/C++, Python, Perl, Tcl, Bash, Java, Go, Assembly (x86)
- **Other:** Git, SVN, Linux, CMake/Ninja, Docker, CI/CD

WORK EXPERIENCE

Cisco, Maynard, MA | ASIC Design Verification Engineer Intern May 2026 - August 2026

- Assisted with verification of 1.6T MACsec IEEE 802.1AE functionality for an optical networking DSP device
 - Debugged failures in a checker by tracing a sampling-rate mismatch between the checker and DUT; confirmed DUT was correct and updated checker, eliminating false failures without masking real defects
 - Redesigned a SystemVerilog functional-coverage model for the block after finding it reported false closure on legal lane configurations; replaced error-prone handwritten bins with a generated approach and validated closure across the full legal configuration space
 - Wrote a datapath reset test covering two reset scopes; traced a post-reset recovery failure to the encryption IP, which required independent reconfiguration, and fixed it with a DPI-based handshake
- Worked on project verification infrastructure support
 - Drove the verification build from 30 compiler errors across 15 files to zero (largest single block ~226 errors), re-enabling strict build and Coverity static-analysis checks in nightly regression, and fixed 5 high-severity defects
 - Triaged ~40 outstanding nightly regression failures across two datapath blocks, automating reproduction from the results database and clearing about a third of the backlog as stale with documented rationale

EarthPulse AI, Barcelona, Spain | Geospatial Data Science Intern May 2025 - July 2025

- Automated satellite data ingestion and preprocessing in Python, reducing pipeline runtime 40% and enabling daily refresh cycles
- Built and deployed pipelines processing 1TB+ of satellite imagery, generating triplet/pair training data for unsupervised super-resolution models

TECHNICAL PROJECTS

[Tclslang \(Tcl Extension for Verilog Analysis\)](#) C/C++, Slang (SystemVerilog), Tcl

- Built a Tcl extension on the slang SystemVerilog compiler that elaborates multi-file designs into a scriptable, EDA-shell-style API (get_module, get_cells, get_ports, get_driver)
- Resolved full instance hierarchies through generate blocks, instance arrays and interface/modport ports, reporting each port's direction, net/variable kind, data type, width and dimensions
- Classified every port connection as net, variable, constant, expression or interface, with source text and data type, and surfaced slang diagnostics as Tcl errors

[Encrypted Concurrent Key-Value Store](#) C++, POSIX, OpenSSL

- Built a multi-client C++ key-value server using two-phase locking, so each in-memory update and its durable log record commit atomically and stay correctly ordered under concurrent writers
- Made writes crash-resilient with direct POSIX I/O (write/fsync) and a custom binary log format, so no committed update is lost if the server crashes

LEADERSHIP & INVOLVEMENT

Computer Science & Business Association, VP of Development / Technical Development Chair (2024–2026)

- Directed the **Lehigh Valley Collegiate Hackathon** (~100 students, six universities); secured **\$10K** in sponsorships